Exp No:19Experiment to understand Logistic Regression for a given

data set.

AIM:

Logistic regression is a supervised machine learning algorithm used to binary classification it predicts the probability of a data point belonging to a particular class.

Description:

Understand the Logistic Regression algorithm for the dataset given.

CODE:

```
import numpy as np

import pandas as pd

df=pd.read_csv('Social_Network_Ads.csv')

df

df.head()

features=df.iloc[:,[2,3]].values

label=df.iloc[:,4].values

features

label

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LogisticRegression

for i in range(1,401):

x_train,x_test,y_train,y_test=train_test_split(features,label,test_size=0.

model=LogisticRegression()

model.fit(x_train,y_train)

train_score=model.score(x_train,y_train)

test_score=model.score(x_test,y_test)

if test_score>train_score:

print("Test {} Train{} Random State {}".format(test_score,train_score,i)

x_train,x_test,y_train,y_test=train_test_split(features,label,test_size=0.2,
```

```
finalModel=LogisticRegression()
finalModel.fit(x_train,y_train)
print(finalModel.score(x_train,y_train))
print(finalModel.score(x_test,y_test))
from sklearn.metrics import classification_report
print(classification_report(label,finalModel.predict(features)))
```

OUTPUT:

1. Original Dataset

	StudyHours	Attendance	Pass
0	1	50	0
1	2	55	0
2	2	60	0
3	3	58	0
4	3	62	0
5	3	65	0
6	4	60	0
7	4	68	1
8	4	70	1
9	5	65	0
10	5	70	1
11	5	72	1
12	6	68	1
13	6	75	1
14	6	78	1
15	7	72	1
16	7	80	1
17	7	82	1
18	8	78	1
19	8	84	1
20	8	86	1
21	9	82	1
22	9	88	1
23	9	90	1
24	10	86	1
25	10	92	1
26	10	94	1

27	11	90	1
28	11	95	1
29	12	96	1

6. Classification Report

	precision	recall	f1-score	support
0	1.00	1.00	1.00	8
1	1.00	1.00	1.00	22
accuracy			1.00	30
macro avg	1.00	1.00	1.00	30
weighted avg	1.00	1.00	1.00	30

2. Features

```
array([[ 1, 50],
       [ 2, 55],
       [ 2, 60],
       [ 3, 58],
       [ 3, 62],
       [ 3, 65],
       [ 4, 60],
       [ 4, 68],
       [ 4, 70],
       [ 5, 65],
       [ 5, 70],
       [ 5, 72],
       [ 6, 68],
       [ 6, 75],
       [ 6, 78],
       [ 7, 72],
       [ 7, 80],
       [ 7, 82],
       [ 8, 78],
       [ 8, 84],
```

```
 [ 8, 86],
 [ 9, 82],
 [ 9, 88],
 [ 9, 90],
 [10, 86],
 [10, 92],
 [10, 94],
 [11, 90],
 [11, 95],
 [12, 96]])
```

3. Labels

```
array([0, 0, 0, 0, 0, 0, 0, 1, 1, 0, 1, 1, 1, 1, 1, 1, 1, 1,
       1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1])
```

4. Final Logistic Regression Model

```
LogisticRegression(max_iter=1000)
```

5. Training and Testing Accuracy

```
Training Accuracy: 1.0
Testing Accuracy: 1.0
```

RESULT:

Thus the logistic regression algorithm was successfully implemented and test to classify the given dataset.